



Model based strategies towards protein A resin lifetime optimization and supervision



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ABSTRACT

The high cost of protein A resins drives the biopharmaceutical industry to maximize its lifetime, which is limited by several processes, usually referred to as resin aging. In this work, two model based strategies are presented, aiming to control and improve the resin lifetime. The first approach, purely statistical, enables qualitative monitoring of the column state and prediction of column performance indicators (e.g. yield, purity and dynamic binding capacity) from chromatographic on-line data (e.g. UV signal). The second one, referred to as hybrid modeling, is based on a lumped kinetic model, which includes two aging parameters fitted on several resin cycling experimental campaigns with varying cleaning procedures (CP). The first aging parameter accounts for binding capacity deterioration (caused by ligand degradation, leaching, and pore occlusion), while the second accounts for a decreased mass transfer rate (mainly caused by fouling). The hybrid model provides important insights into the prevailing aging mechanism as a function of the different CPs. In addition, it can be applied to model based CP optimization and yield forecasting with the capability of state estimation corrections based on on-line process information. Both approaches show promising results, which could help to significantly extend the resin lifetime. This comes along with increased understanding, reduced experimental effort, decreased cost of goods, and improved process robustness.

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1. Introduction

Protein A affinity resins are the most widely applied stationary phases for the capture of monoclonal antibodies (mAbs). However, their high selectivity and impurity clearance come with high cost [1–4]. Protein A resins are nearly 50% more expensive than non-protein-based stationary phases and manufacturing-scale columns are in the order of one million dollars per packing [5,6]. These high investments demand careful utilization, which can be achieved using continuous chromatography as well as maximizing the resin lifetime [7]. Resin aging results in performance loss and limits the lifetime usually to around 50–100 cycles [5]. Hence, an active process-monitoring procedure is needed that tracks column performance indicators, such as yield, dynamic binding capacity (DBC),

product quality, concentration (UV) profiles during loading, elution, and regeneration or the height equivalent to a theoretical plate (HETP). Impurity content of elution fractions, such as aggregates, residual host cell proteins (HCP), host cell DNA and leached protein A (LPA), are also routinely analyzed off-line [5,8,9].

Resin aging is a complex phenomenon and a detailed fundamental understanding is still missing. However, the manifold causes can be roughly attributed to two mechanisms. On one hand, there are processes that affect the resin capacity, including ligand and leaching, degradation and inactivation, resulting in decreased binding capacity. On the other hand, there are processes related to mass transfer, including packed bed and resin fouling as well as degradation and pore blocking of the chromatographic support [10–14].

A lot of effort has been spent on maximizing resin lifetime. At the material structure level, an important approach is the improvement of the backbones towards reduced compressibility and increased mass transfer rates allowing faster cycling, as well as

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optimization of the ligand chemical resistance [4,15,16]. On the process level, a crucial optimization potential lies in the development of suitable cleaning procedures (CP). The optimal CP should be sufficiently harsh to eliminate resin fouling while avoiding ligand degradation. Mechanistic investigations by Jiang and coworkers [14] showed that, in the specific case they considered, the primary factor for resin aging was ligand degradation due to the very harsh strip and regeneration conditions. In other studies, it was concluded that protein built-up in the intraparticle pores resulted in a decreased access to the ligands, which was further confirmed by transmission electron microscopy [13,17]. These different conclusions show that the optimum operation in general depends strongly on the combination of the used chromatographic process, the considered supernatant and the stationary phase. The CP development usually requires very laborious resin lifetime studies at lab scale. Here, the maximum number of cycles as well as the optimized column regeneration and storage conditions are determined and later validated at manufacturing scale. However, fixing the number of cycles at manufacturing scale might bear considerable risks, due to resin lot-to-lot variability, differences in feed material, column storage, packing quality or other unforeseen events [18–21]. This risk is typically mitigated by adding a safety margin, however at the cost of higher production costs.

In such a situation, further optimization potential lies in on-line monitoring and control of column performance. This might enable the definition of a flexible resin lifetime, to be adjusted on-line, so that more cycles than the proposed cycle number arising from the a-priori laboratory resin lifetime studies can be performed. Moreover, additional column repacking or cleaning steps could be suggested through the gained understanding. Several examples showed already the potential of using statistical tools such as principal component analysis (PCA) to infer on integrity breaches and visualize the column quality by analyzing the transitions within the chromatograms [8,21,22]. Examples of deterministic models are rather scarce, probably because of the limited availability of suitable data and the complex phenomena involved in resin aging. Behere et al. [23] developed a modified shrinking core model, which described the lumped ligand degradation including protein A leaching, unfolding, and degeneration. Although the model could well predict the effects of caustic on resin performance, it did not include the impact of fouling, since purified feed was used. Shekhawat et al. [24] described the binding and mass transfer characteristics through a simplified linear driving force and kinetic model. A fouling factor, function of the binding and mass transfer characteristics of the resin and of its regeneration was empirically introduced using a combination of Buckingham's π theorem and statistical analysis. The model described the protein A capture performance well and improved the understanding of the underlying mechanisms. However, this work did not specifically include the ligand degradation and leaching as a function of the CP.

In this work, we present an innovative procedure, based on two different model approaches, aiming to improve and supervise the resin lifetime. The first approach is based on statistical techniques, such as principal component analysis (PCA) and partial least squares (PLS) regression, and includes the analysis of an appropriate dataset of lifetime studies of a continuous chromatographic unit. The second approach is based on a hybrid model including a deterministic lumped kinetic model with two adjustable aging parameters. These account for the decrease in binding capacity and mass transfer, respectively, in dependence of the resin cycling. The parameters are fitted on four column lifetime experimental data sets, where the CP harshness was varied, thus resulting in different aging behaviors. The advantages and weaknesses of both approaches are discussed in connection with their industrial application. The benefits range from an increased understanding of the resin aging mechanism and model based CP design, to

advanced monitoring of several column performance parameters with on-line updated forecasting capability leading to better policies for resin utilization.

2. Materials and methods

2.1. Materials, equipment and experimental methods

One resin lifetime cycling study was performed on a CUBE Combined (ChromaCon® AG, Zurich, Switzerland) and four cycling studies were performed on ÄKTA Avant 150 systems (GE Healthcare, Uppsala, Sweden). The resulting datasets are referred to as, CUBE-DS and ÄKTA-DS1 to 4, respectively. In the former, the study was performed using the CaptureSMB procedure, a continuous mode that uses two columns operated in series to improve resin utilization [25,26], while in the others the classical batch single mode operation was adopted. The absorbance at 280 nm, pressure drop, conductivity, and pH were recorded for studies with both the ÄKTA Avant and CUBE systems. Cell culture supernatant containing a recombinant monoclonal antibody (mAb) with concentrations between 4.1 and 5.7 mg/mL was used as feed material, including high contents of impurities. Within each cycling lifetime study the same supernatant was used, thus ensuring a constant product to impurity ratio. MabSelect SuRe LX resin (GE Healthcare, Uppsala, Sweden) was used and packed in 1 cm inner diameter Omnifit column hardware (Kinesis Inc., West Berlin, NJ) with bed heights specified in Table 1.

Each purification cycle consisted of equilibration, loading, three wash steps, elution, cleaning in place CIP 1, neutralization 1, CIP 2 and neutralization 2. The duration and buffer conditions of each step were constant between different cycling studies, except for the CIP and neutralization steps. The buffers used for CIP were based on NaOH, varying in concentrations within the manufacturer recommendations. For quantitative modeling purposes, a harshness factor (HF) is introduced as a measure of the severity of the cleaning procedure, including both CIP 1 and CIP2, as follows:

$$HF = \frac{c_{\text{buffer}} \cdot t_c}{M_{\text{resin}} \cdot \omega} \quad (1)$$

where c_{buffer} is the CIP buffer concentration, t_c the total resin contact time, M_{resin} the resin mass and ω a normalizing constant. The HF values relative to each of the datasets mentioned above are summarized in Table 1. It can be seen that the cleaning procedures used in CUBE-DS1 and ÄKTA-DS1 show the lowest HF value of 50, while the one used in ÄKTA-DS2 appears as the most severe one with the highest HF of 500.

The mAb concentrations of the column loads and effluents were determined by a Protein-A UPLC system (Waters Corp., USA) and UV absorbance at 280 nm with a DropSense96 UV-vis spectrophotometer (Trinean, Belgium), respectively. Yield was calculated by dividing the protein amount in the elution fraction with the total loaded protein amount. In ÄKTA-DS2 - DS4 the yield was measured for each cycle. However, in CUBE-DS1 the yield was only measured from pooled cycles (i.e., cycle 1–14, 15–20, 21–28, 29–36, 37–44, 45–52 and 53–60). This was also the case in the ÄKTA-DS1, where the same pools as in CUBE-DS1 plus those of cycles 61–65 and 66–70 were analyzed. Hence for the latter two data sets, the missing yield values were linearly interpolated for each cycle.

Breakthrough curves (BTC) were performed with the ÄKTA Avant 150 system by frontal loading of PBS diluted Protein A eluate after selected cycles, applying feed concentrations and flow rates as described in Table SI 1. From the obtained BTC, the DBC₁₀ and DBC₇₀ values were determined at 10% and 70% breakthrough, respectively.

Additionally, the elution fractions of the CUBE-DS were analyzed for impurities. ELISA for quantification of residual Chinese

Table 1
Experimental conditions of the cycling lifetime studies.

	Column length [cm]:	Load flow rate [mL/min]:	Load amount [mg]:	Feed conc. [mg/mL]:	HF:	Performed cycles:
CUBE-DS	2 × 10	0.36 ¹ / 1.96 ²	510	4.1	50	60
ÅKTA-DS1	10.5	0.98	412	4.1	50	70
ÅKTA-DS2	20	1.75	785	5.7	500	75
ÅKTA-DS3	20	1.75	785	4.6	250	100
ÅKTA-DS4	20	1.75	785	4.6	83	65

Loading flow rate during batch phase¹; Loading flow rate during interconnected phase²

hamster ovary (CHO) HCP and residual LPA was conducted in high throughput mode using the Tecan liquid handling system. HCP levels were quantified using the CHO HCP 3rd generation kit (Cat # F550, Cygnus Technologies, Southport, NC), while LPA was quantified using Repligen Protein-A ELISA kit (Cat # 9000-1, Repligen Corporation, USA) according to manufacturer's protocol. Absorbance was measured at 450/650 nm using EnVision Multilabel reader (PerkinElmer, Waltham, MA). Data quantification and analysis was carried out using JMP software (SAS Institute Inc. Cary, NC). Residual CHO DNA in the samples was measured using real-time quantitative PCR (RT-qPCR). RT-qPCR was carried out with the 2 × TaqMan Universal PCR Master Mix kit (Life Technologies, USA) and 7900 HT Real Time PCR system (Life Technologies, USA) according to the manufacturer's instructions. These analytics were performed on every elution pool and interpolated for each cycle.

2.2. Computational methods of the statistical approach

In both approaches the calculations were performed using MATLAB R2018a (Mathworks, Natick, MA) and in-house developed routines, if not stated otherwise.

2.2.1. Data preparation

For the statistical model approach only the CUBE data set and the corresponding column performance indicators, including yield, DBC₁₀, DBC₇₀, LPA, DNA and HCP, were used. The general data pretreatment procedure included the following steps:

- 1 Chromatographic phase separation & horizontal alignment. Based on operating conditions, the chromatograms of each cycle were divided into 12 different phases, which enabled the removal of redundant information (e.g. phases with plateaus) to avoid sensitivity reduction in the analysis and modeling steps [8]. The horizontal alignment was achieved through the conversion of the data into the column volume dimension, which was determined after each column repacking. Since phase length differences between different cycles were negligible, no data shifting algorithms were applied. The small differences in data points were compensated by cutting uninformative points at the end of each phase. As an illustrative example, the chromatographic raw data of UV1 is shown for all phases in Figures SI 1-2.
- 2 Spike removal & smoothing. The CUBE raw data showed several exogenous spikes in the UV, conductivity, pH and pressure signal. An algorithm was developed, which automatically detected and removed spikes by replacing them with averaged neighboring values, as shown in Figure SI 3 (A-B). Subsequently, the Savitzky-Golay algorithm [27] was applied to smooth the noisy pressure signal by using a second polynomial order, a window size of 2% of the phase length without a derivative. The effect of the smoothing on a selected pressure profile is shown in Figure SI 3 (C).
- 3 Unfolding, centering & scaling. Prior to the next pretreatment steps, the three-way chromatographic data tables with N cycles, J time points and K variables were batch-wise unfolded into a

two-dimensional matrix X with N rows and J × K columns [28]. The column performance parameters (e.g. yield, DBC and impurities) of each cycle were arranged in a Y matrix, corresponding to the rows of the X matrix. Subsequently, Standard Normal Variate (SNV) processing was applied on the row level of the X matrix [29]. Additionally, either mean-centering (MC) or auto-scaling (AS) was applied column-wise on the X and Y matrices [30].

2.2.2. Data analysis and modeling techniques

PCA was applied to investigate patterns, groups and evolutions of the cycles from the CUBE-DS. The MATLAB built-in function *pca* was used to analyze the pretreated X matrices including different phases and process variables, as indicated for each analysis. The results were visualized by score and loading plots [31,32]. PLS regression was used to regress the X matrix on Y matrix, using the NIPALS algorithm, including a 5-fold cross-validation (CV) [33,34]. The optimal number of latent variables (LV) was determined based on the minimum error in CV. In order to evaluate the model performance, its root mean square error (RMSE) was calculated as follows:

$$RMSE = \sqrt{\frac{1}{n} \cdot \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

Where $\hat{y}_i$ is the predicted value of observation i , y_i is the reference value of observation i , and n the total number of observations. While the RMSECV indicates the error from the CV on data from CUBE-DS column 1, the RMSEP indicates the prediction results on CUBE-DS column 2. The relative RMSE (relRMSE) was calculated by dividing the RMSE with the standard deviation of the reference values. Moreover, the coefficient of determination (R^2) was calculated for the prediction on column 2 as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

The MATLAB built-in function *bayesopt* was used to select the best combination of process phases and process variables for X with the objective of minimizing the prediction error for each column performance indicator [35].

2.3. Computational methods of the hybrid approach

2.3.1. Data preparation

The hybrid model was fitted on the BTCs measured on the ÅKTA units. Within the Kalman filter approach (KF), the UV signal of the loading and elution curves were used for model updating. Hence, the pretreatment only includes the phase separation to extract the loading and elution profiles. A calibration curve was used to transfer the UV signal into mAb concentrations.

2.3.2. Model description

In this work, the lumped kinetic model [36,37] was chosen as the backbone for the hybrid model. The model can be formulated

as follows:

$$\frac{dc}{dt} = -v \frac{dc}{dx} + D_L \frac{d^2c}{dx^2} - \varphi \frac{dq}{dt} \quad (4)$$

$$\frac{dq}{dt} = k_m (q^* - q) \quad (5)$$

$$v = \frac{Q}{A_{col} \varepsilon} \quad (6)$$

$$\varphi = \frac{(1 - \varepsilon)}{\varepsilon} \quad (7)$$

$$D_L = A \frac{d_p}{2} v \quad (8)$$

$$q^* = \frac{H c}{1 + \frac{H c}{q_{sat}^0}} \quad (9)$$

$$k_m = k_{max}^0 \left(S_1 + (1 - S_1) \left(1 - \frac{q}{q_{sat}^0} \right)^{S_2} \right) \quad (10)$$

The protein product mass balances in both the liquid and the adsorbed phase, are presented as partial differential equation along the column longitudinal axis x in Eq. (4)–(5), where c and q describe the liquid and solid phase concentration of the protein, respectively, k_m is the mass transfer coefficient and q^* the equilibrium solid phase concentration of the protein. The interstitial velocity v and the phase ratio φ are defined in Eq. (6)–(7), where Q is the volumetric flow rate, A_{col} the column cross sectional area and ε the bed porosity. The axial dispersion and the Langmuir isotherm are modeled using Eq. (8)–(9). Here, D_L represents the apparent axial dispersion coefficient, A the intercept of the reduced van Deemter equation and d_p the average particle diameter. Furthermore, H is the Henry coefficient, q_{sat}^0 is the maximum saturation capacity of the resin, k_{max}^0 is the maximum mass transfer coefficient, S_1 the maximum hindrance coefficient and S_2 accounts for the non-linear increase of the hindrance.

The system of Eq. (4)–(10) can be used to describe the bind and elute operation during a cycle of the chromatographic column. The parameters $H, q_{sat}^0, k_{max}^0, S_1, S_2, \varepsilon$ and A were fitted on the BTC, which was performed at the beginning of the lifetime studies, and referred to as *fixed* or *time invariant* parameters. In order to introduce the effect of resin aging, a hybrid modeling approach is taken [38], in which the following equations are introduced:

$$q_{sat} = q_{sat}^0 \theta_1 \quad (11)$$

$$k_{max} = k_{max}^0 \theta_2 \quad (12)$$

The cycle dependent parameters θ_1 and θ_2 represent the aging of the resin, which reduce the maximum saturation capacity and the mass transfer coefficients, respectively, and will be referred to as *aging parameters*. These can be correlated with the *HF* and the protein loading *PL* using an empirical model given as follows:

$$\theta_1 = f_1(HF, PL) \quad (13)$$

$$\theta_2 = f_2(HF, PL) \quad (14)$$

where *PL* accounts for the protein mass loaded to the column in each cycle, divided by the volume of the column. By including these equations, the model is able to predict the performance of the column at different cycles during its lifetime, according to the loading and the cleaning strategy used. The data-driven part could be of course extended to additional process or resin specific variables, through the addition of appropriate experimental data.

Finally, to simulate the complete system, the model has to be integrated with proper process variables and initial conditions. Protein is fed to the column with constant inlet concentration (harvest pool concentration) and a constant flowrate. Accordingly, the following initial and boundary conditions are applied:

$$c_{(t=0,x)} = c_{0(x)} \quad (15)$$

$$q_{(t=0,x)} = q_{0(x)} \quad (16)$$

$$\left. \frac{dc}{dx} \right|_{x=L_{col}} = 0 \quad (17)$$

$$c_{(t,x=0)} = c_{in} + \frac{D_L}{v} \left. \frac{dc}{dx} \right|_{x=0} \quad (18)$$

At the beginning of the operation, the protein concentration in the liquid and solid phase are zero at every point of the column, thus both $c_{0(x)}$ and $q_{0(x)}$ are zero. Besides different operation modes, the model and assumptions for the BTC experiments and the bind-elute operation for the cycling study are identical. The MATLAB built-in function *ode15s* was used for the numerical solution of the model equations.

2.3.3. Model parameter fitting

In order to fit the model, the following inverse problem has to be solved

$$\min_p Er = \sum_i (y_{(p)i} - y_i^{exp})^2 \quad (19)$$

subject to

$$y = h(p, u) \quad (20)$$

$$p_{min} \leq p \leq p_{max} \quad (21)$$

Here, y_i^{exp} represents the experimental outlet concentration c at time i , while y is the calculated outlet concentration using the system of equations h (Eq. (4)–(10)), where p is the vector of model parameters and u is the vector of process conditions. This problem can be independently solved for every BTC of the dataset to get a set of model parameters, but most of the parameters (namely *fixed parameters*) are kept constant along the lifetime of the resin.

In order to correlate the aging parameters to the cleaning and loading conditions, two different approaches involving Gaussian processes (GP) were applied. In machine learning, a Gaussian GP is a nonparametric model, which has the powerful characteristic of giving the probability distribution of the prediction, based on similarity with a training dataset [39]. Once the GP is obtained from the training set, it can be used to predict both the mean and standard deviation of the output variable under known input conditions.

In the first approach, the θ parameters are used to fit two GPs (one per θ) with the accumulated *PL* and the accumulated *HF* as the input variables, which are reported in Table 4:

$$\theta = GP(HF_{Acc}, PL_{Acc}) \quad (22)$$

The accumulated *HF* at a given cycle N is defined as $HF_{Acc_N} = \sum_{i=0}^{N-1} HF_i$, where HF_i is the harshness factor of each cycle i . The same definition applies for PL_{Acc} .

With these models for θ_1 and θ_2 , the aging of the resin (expressed as aging parameters) can be correlated with the accumulated *HF* and *PL*, which are related to the cleaning and loading of the column from the beginning of its lifetime.

The second approach is to fit the two GPs with the changes in the θ parameters with regards to the amount of protein and the harshness factor of each cycle, and the current value of θ (θ_{cur}). This is addressed as follows:

$$\Delta\theta = GP(\theta, HF, PL) \quad (23)$$

Then, the θ parameter for the next cycle is calculated as:

$$\theta_{cur+1} = \theta_{cur} + \Delta\theta \quad (24)$$

The second approach can be integrated into a dynamic simulation of the system, as required for the KF. Both approaches are applied in Section 3.2.1-2 and Section 3.2.3, respectively. The fitting

problem was solved by using *fmincon* from MATLAB's Optimization Toolbox. All GP implementations were based on *fitrgp* from MATLAB's Statistical and Machine Learning Toolbox.

2.3.4. Cleaning strategy design and model updating

The hybrid model was used to optimize the cleaning strategy in terms of the optimal *HF* in dependence of a certain *PL* to achieve a maximum average yield for a fixed number of cycles. The mathematical problem that describes the process optimization can be formulated as follows:

$$\max_u J \quad (25)$$

subject to

$$J = f'(p, u) \quad (26)$$

$$u_{\min} \leq u \leq u_{\max} \quad (27)$$

Here, J represents the objective function to be optimized, f' is the hybrid model and u is the vector of process variables to be optimized. It is worth noting that f' include the mechanistic model described in Eq. (4)–(18) (with its parameters vector p) along with the GP model described in Eq. (22). In this work, J was chosen to be the average yield for 100 cycles, while u represents the *HF* (the *PL* was fixed). Other problems can be solved as well, including different process variables and objective functions, such as the number of cycles with a given yield.

The KF was applied to update the predictions of the hybrid model with the on-line yield derived from the UV signal of the loading and elution curve. The implementation is similar to the one described by Feidl et al [40].

3. Results and discussion

3.1. Statistical approach

In the following sections, two statistical approaches are applied to infer the column quality. Within the first approach, chromatographic on-line signals, such as UV, conductivity and pressure, serves as input for a PCA to qualitatively monitor the column state. In the second approach, the correlations between chromatographic

data and column performance indicators are analyzed with PLS to quantitatively monitor the column quality.

3.1.1. Qualitative monitoring of column quality

PCA was performed on data from the elution phase of column 1 from the CUBE-DS, including the UV, conductivity, and pressure, since the elution phase and the corresponding process parameters showed the most significant changes with resin reuse. The PCA was build on only those cycles having yield larger than 80% (i.e. cycle 1–47). Subsequently, the data were batch-wise unfolded and column-wise auto-scaled, resulting in an X matrix with 47 rows (number of cycles) and 3348 columns (datapoints of elution phase $\times$ process variables). In Fig. 1, the resulting scree plot of the PCA is shown, where the explained variance (bars) and cumulated explained variance (curve) for each principal component (PC) are shown as a function of the number of PCs. It can be observed that PC1 accounts for 35% of the explained variance, while PC2, PC3, and PC4 explain 26%, 19%, and 8% respectively. This means that 88% of the variance in the 3348 dimensional dataset can be explained by only four PCs, which confirms the significant dimension reduction capability of PCA.

This enables the efficient illustration of the complex data set within a reduced dimensional space. The corresponding score plot including PC1 and PC2 is shown in Fig. 2. Each colored point represents one of the cycles on which the PCA was built. Additionally, a yield based confidence ellipse was calculated within the multivariate space, based on cycles with a yield above 85%, and plotted into the score plot. If the corresponding score (cycle) shows up within the ellipse, there is a 97.5% confidence that the cycle has a yield $> 85\%$.

Mainly two phenomena can be observed in the score plot. First, there is a clear tendency to move from the upper left corner to the lower right as the number of cycles increases. In particular, one can identify three groups of cycles: 1 to 20, 21 to 36 and 37 to 47, each exhibiting such a trend, where at the higher cycles positive values of PC1 and negative values of PC2 are actually reached. However, after the two column repackings (i.e., after cycles 20 and 36), the system moves back to the upper left area. The corresponding loading plots, shown in Figure SI 4 (A–C), indicate that the cycles are

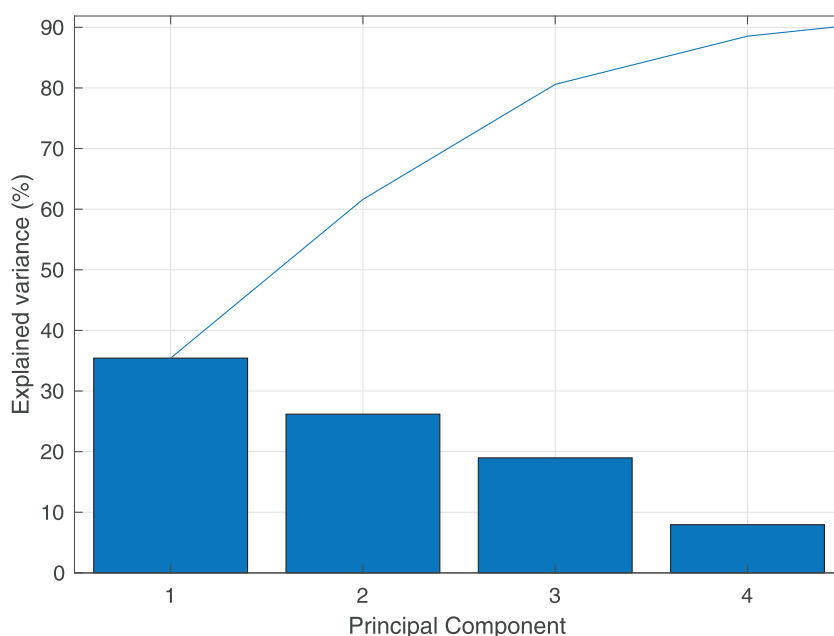


Fig. 1. Scree plot derived from the PCA of the elution phase of the CUBE-DS, including the UV, conductivity and pressure signal. Bars indicate the explained variance for the first four principal components, while the line shows the cumulated explained variance.

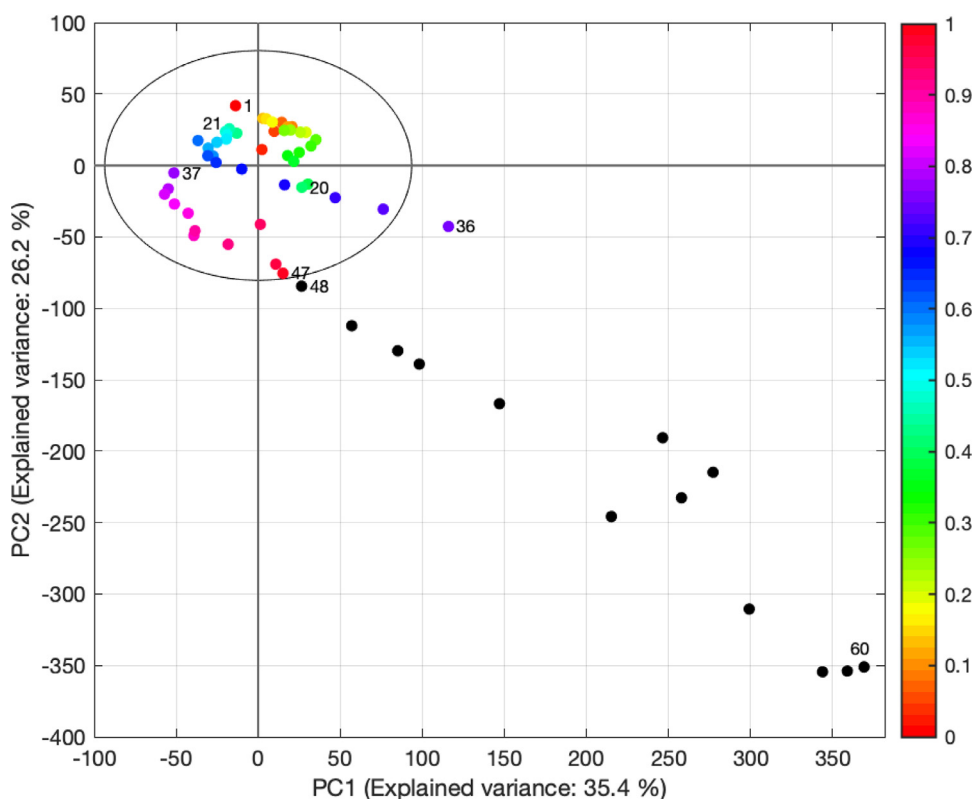


Fig. 2. Score plot of PC1 and PC2 derived from the PCA of the elution phase of the CUBE-DS, including the UV, conductivity and pressure signal. Each point represents one cycle with the respective cycle number. PCA was built on colored cycles (yield > 85%), while black cycles were projected in the reduced data space. Ellipse indicates 97.5% confidence to be within the 85% yield specification.

mostly shifted in the direction of PC1 by the increase of pressure in the column, while mainly a decrease of the UV elution curve shifts the cycles to the negative direction of PC2. Although the conductivity shows a clear contribution to both PCs, its character is more difficult to interpret.

Those findings are typical of resin aging. The observed increase in pressure as well as the broadening of the UV elution peak (shown in Figure SI 1 D) are most probably due to resin fouling, while the decrease in the height of the UV elution peak relates more to the decrease in binding capacity due to leaching and/or pore occlusion. Changes in conductivity profiles during buffer transitions might be an effect of alterations in the packed bed quality. This is coherent with the observation that column repacking significantly reduced the shift along the PC1, but could not eliminate the shift towards the negative direction of PC2. The last 13 cycles (48 to 60), showing a low yield, were projected into the existing PC space and are represented by the black circles in Fig. 2. It is no surprise that all of those cycles are clearly out of the specification ellipse and show an even stronger tendency to move towards the lower right corner.

These results show how PCA can be applied as a process monitoring tool. Ideally, it is trained on a large historical data set, which includes many cycles that are within the normal acceptable range, and it is applied by projecting on-line data into the same space. This visualizes whether the current cycle is still within the normal acceptable range (i.e., the ellipse in this case) and, once cycles are projected close to this border, actions such as column repacking or an additional column cleaning can be initiated. In this case, it appears that an additional repacking and cleaning step after cycle 47 would have brought the process back inside the ellipse and further cycles could have been performed while remaining within the acceptable range of yield values.

3.1.2. Quantitative predictions of column performance

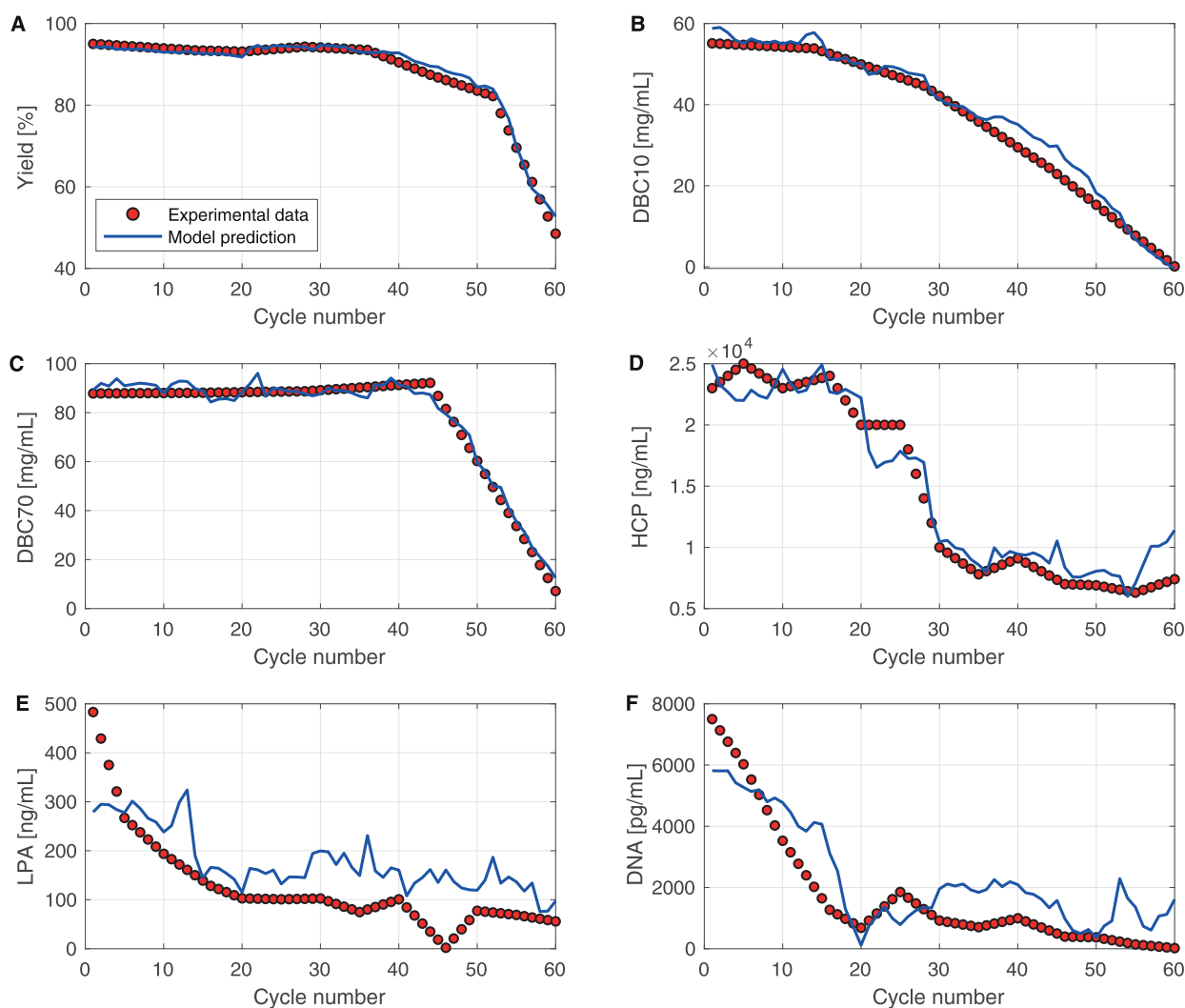
PLS regression was used to quantitatively predict chromatographic performance parameters such as yield, DBC₁₀, DBC₇₀, LPA, DNA and HCP based on the on-line signals discussed above. The models were trained on data of column 1 from CUBE-DS and validated on data of column 2 from CUBE-DS. The optimal combination of phases, variables and pretreatments used as PLS data inputs were determined based on Bayesian optimization and are summarized for each predicted performance parameter in Table 2. Additionally, the prediction range, optimal number of latent variables, different RMSE as well as R² are reported.

It is found that, in general, the wash step (phases 2 and 3), elution step (phase 4), and CIP steps (phases 5 and 6) were most often selected by the optimizer, as well as the UV absorbance, conductivity, and pH, highlighting their high information content, which is needed to describe the aging related decay of performance indicators. It can be seen that yield, DBC₁₀, and DBC₇₀ show very low reRMSEP in the range 0.13 to 0.16, while the quality related predictions of HCP, DNA and monomer showed slightly larger reRMSEP in the range 0.29 to 0.65, which can be still considered as an acceptable semi-quantitative model. The predictions of LPA showed a very high reRMSEP of 1.57 and a negative R² of -0.46, which clearly indicates that in this case the PLS model is not statistically significant.

Besides investigating the error metrics, the model performance can be analysed by visually comparing the predictions (blue curve) and the offline measured and interpolated reference values (red circles) shown in Fig. 3 (A-F) as a function of the cycle number. It can be seen that the predictions match nicely the observed yield, DBC₁₀, and DBC₇₀. Although the predictions of the HCP and DNA contents are noisier, they correctly follow the measured trends as well. This is particularly true for HCP (Fig. 3D) and, to some ex-

Table 2PLS prediction results of the yield, DBC₁₀, DBC₇₀, HCP, LPA and DNA.

Predicted variable	Phases	Variables	Pretreat-ment	Pred. range	Opt. LV	RMSECV	relRMSECV	RMSEP	relRMSEP	R ²
Yield [%]	2, 4, 5	UV, conductivity	AS	48–95	5	1.29	0.12	1.43	0.13	0.98
DBC ₁₀ [mg/mL]	4, 5, 6	UV, pH	MC	0–55	5	1.86	0.11	2.47	0.15	0.98
DBC ₇₀ [mg/mL]	2, 3, 4	UV, conductivity, pH	MC	7–92	8	3.06	0.16	3.15	0.16	0.98
HCP [ng/mL]	2, 3, 5	UV, conductivity, pH	MC	6300–25000	4	2142	0.36	1728	0.29	0.95
LPA [ng/mL]	2, 3, 6	Pressure, pH	AS	2–483	9	43	0.60	113	1.57	-0.46
DNA [pg/mL]	3, 5, 6	UV, pH	AS	20–7500	7	644	0.39	1062	0.65	0.70

**Fig. 3.** PLS predictions (lines) and reference measurements (dots) of each cycle from CUBE-DS for (A) yield, (B) DBC₁₀, (C) DBC₇₀, (D) HCP, (E) LPA, (F) DNA

tent, for DNA (Fig. 3F), where the model is correctly predicting the initial decrease in DNA content. As mentioned above, the model is instead unable to explain the observed LPA values. In general the analytical standard deviations of the HCP, DNA, and LPA assays are higher compared to the UPLC based measurements. Such analytical differences, but also the low sensitivity of such assays, could be certainly a reason for the deviations of the prediction results.

It is worth mentioning that the statistical model approach has been also applied to the ÄKTA-DS and showed promising results when applied within the same column. However, the transfer to another ÄKTA-DS lead to poor results (data not shown). This can be explained by the differences in CP, which affect the correlations between on-line measurements and performance indicators, thus limiting the extrapolation capability of this approach.

Of course, to fully validate such tools more data are required, which are generally available at the pilot and manufacturing stage. However, from the results above we can certainly conclude that the PLS predictions of the column performance indicators provide a strong quantitative complement to the PCA: the general qualitative state of the column can be followed by PCA while specific quantitative performance indicators can be monitored by PLS.

3.2. Hybrid approach

In the following sections a hybrid model is developed, describing the resin aging as a function of the harshness factor, HF of the used cleaning strategy and amount of protein loaded, PL. The resulting hybrid model is then applied to infer on underlying aging

Table 3Values of the *fixed parameters* fitted on the BTC1 of ÄKTA-DS1.

Parameter	Symbol	Unit	Value	Standard deviation
Henry coefficient	H	-	155.5	7.24
Maximum saturation capacity	q_{sat}^0	mg/mL resin	175.92	2.38
Maximum mass transfer capacity	k_{max}^0	s ⁻¹	0.0031	2.02E-04
Pore blockage coefficient 1	S_1	-	0.3091	1.71E-02
Pore blockage coefficient 2	S_2	-	4.948	4.93E-02
Bed porosity	ε	-	0.386	6.41E-03
Intercept of van Deemter curve	A	-	99.95	4.36

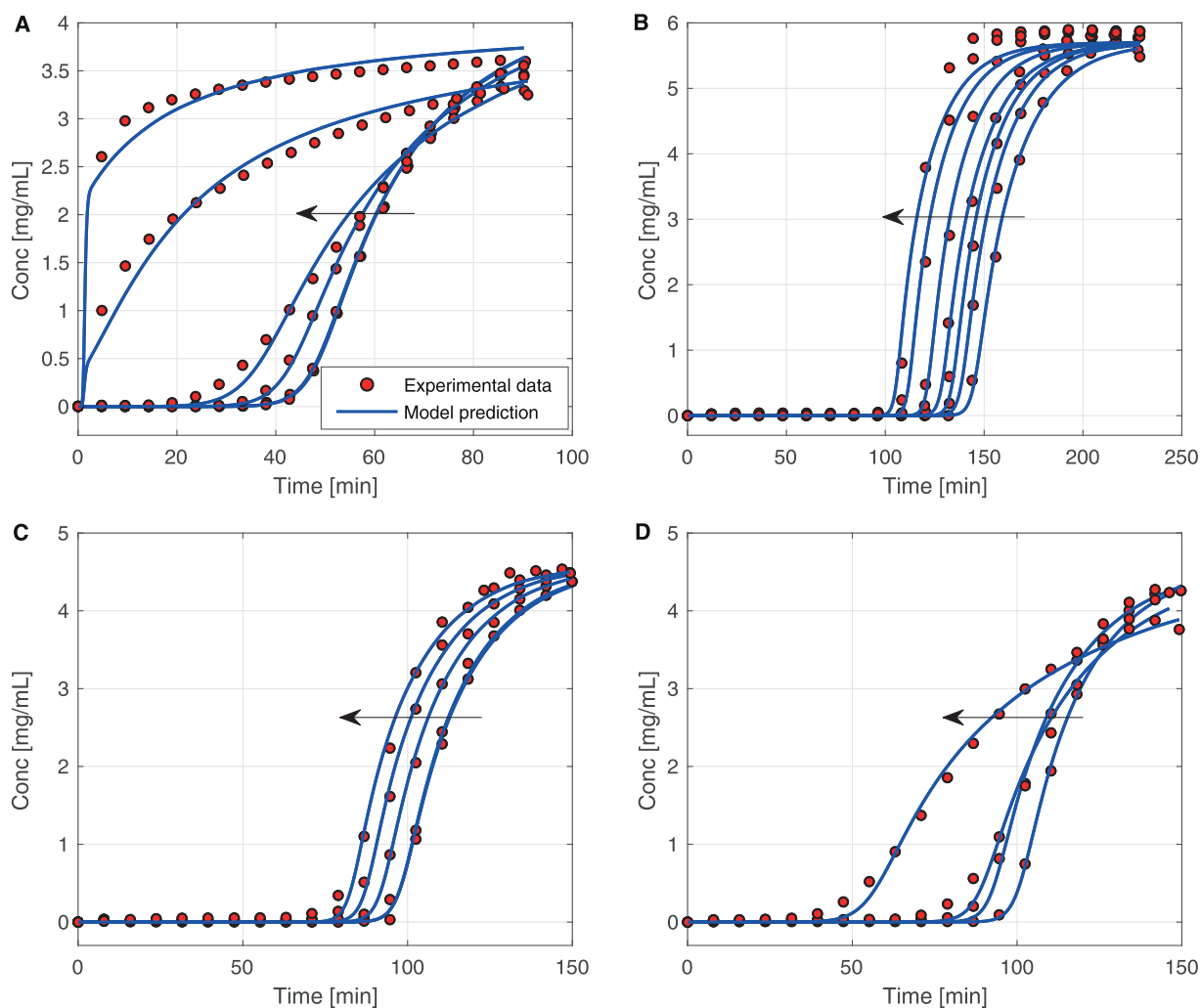


Fig. 4. Comparison of model predictions after θ_1 and θ_2 fitting (lines) and experimental BTCs (dots) for (A) ÄKTA-DS1, (B) ÄKTA-DS2, (C) ÄKTA-DS3, (D) ÄKTA-DS4. The arrows indicate increasing cycle numbers.

mechanisms, for model based design of the cleaning strategy, and for the supervision of the resin state in the frame of an on-line maintenance approach.

3.2.1. Analysis and understanding of the aging process

As discussed above, the *fixed parameters* of the underlying mechanistic model were fitted on the first BTC of ÄKTA-DS1 (performed after the first cycle), and assumed to be constant throughout the following BTCs (while both *aging parameters* θ were kept equal to one). The obtained values for the *fixed parameters* are summarized in Table 3.

Subsequently, the effect of aging was accounted for by fitting the parameters θ_1 and θ_2 , on all the remaining BTCs measured at different cycles of a certain ÄKTA-DS, corresponding to a specific

cleaning procedure. The obtained results are shown in Fig. 4 for the four considered CPs. While the lines represent the fit of the model, the dots indicate the experimental data of the BTC. It can be observed that the model describes the BTC well for each data set, indicating a successful parameter fitting. Additionally, it is worth noting that the four different sets of BTCs exhibit quite different patterns, as a consequence of the different harshness factor characterizing the different CPs. Especially in the case of ÄKTA-DS1, it is seen that the last two BTCs are very steep and show immediate breakthrough, indicating very strong resin aging.

The *aging parameters* θ_1 and θ_2 , the corresponding accumulated protein loading (PL) and harshness factor (HF), as well as the HF/PL ratio, are summarized for each ÄKTA-DS in Table 4. This gives an interesting insight and fundamental understanding of the effect of

Table 4

Cumulated loaded protein amount (PL), cumulated cleaning procedure harshness factor (HF), HF/PL ratio, yield as well as the aging parameter θ_1 and θ_2 for each BTC (performed after the indicated cycle), for all considered ÄKTA-DS.

DATASET	BTC	PL_{Acc}	HF_{Acc}	HF/PL	Yield [%]	θ_1	θ_2
ÄKTA-DS1	1	50	50	1.00	95	1.00	0.98
	14	700	700		95	1.00	0.96
	28	1399	1399		94	0.97	0.75
	44	2198	2198		87	0.97	0.52
	60	2998	2998		52	0.97	0.08
ÄKTA-DS2	70	3498	3498	10.00	21	0.40	0.08
	1	50	500		94	1.00	1.00
	10	500	4997		81	0.82	1.00
	20	999	9993		83	0.77	1.00
	30	1499	14990		80	0.74	1.00
	40	1999	19986		75	0.71	1.00
	50	2499	24983		64	0.66	1.00
	65	3248	32478		55	0.58	0.99
	75	3748	37474		46	0.58	0.99
	100	4997	24983		77	0.76	0.92
ÄKTA-DS3	1	50	250	5.00	95	1.00	1.00
	25	1249	6246		96	0.92	0.93
	50	2499	12491		86	0.86	0.93
	75	3748	18737		83	0.81	0.93
ÄKTA-DS4	100	4997	24983	1.67	77	0.76	0.92
	1	50	83		96	1.00	1.00
	25	1249	2082		94	0.88	0.84
	50	2499	4164		86	0.88	0.67
	65	3248	5413		67	0.79	0.28

the CP and the loaded amount of protein on the aging factors. The HF/PL ratio is an indicator of the harshness of the CIP strategy per gram of protein loaded. For ÄKTA-DS1, the HF/PL ratio is the smallest among all datasets, implying mild cleaning conditions. The corresponding values of θ_1 , accounting for the decrease in binding capacity, remain high for the first 60 cycles and sharply decrease afterwards. The values of θ_2 , which describe the worsening in mass transfer, start much sooner to decrease down to a very low value of 0.08. This agrees with the expectation that, in this case, given the mild CP conditions, the primary aging factor is fouling, reducing significantly the mass transfer and subsequently decreasing the binding capacity due to pore occlusion.

For ÄKTA-DS2, the CP is the harshest among all the studies reported here, having the highest HF/PL ratio. Here, the changes in the aging parameters are opposite to the one in ÄKTA-DS1. While θ_1 strongly drops to 0.58, θ_2 stays constant without a significant decline. This suggests that the mass transfer is not hindered, but the binding capacity is significantly decreased. This can be explained considering that the strong cleaning strategy prevents the formation of fouling, but at the expense of protein A degradation.

For ÄKTA-DS3 a similar effect can be seen, but since the HF/PL ratio is lower, the drop of θ_1 is weaker. This is translated into a longer life of the resin, still affected by degradation, but to a lesser extent than in the case of ÄKTA-DS2. Finally, ÄKTA-DS4 has a similar behavior to ÄKTA-DS1 regarding the formation of fouling, but because of a slightly harsher CP the column degradation is slower. Based on these results, it can be concluded that there should be an intermediate value of the HF/PL ratio that optimizes process performance, such as the maximum number of cycles (for a given yield threshold) or the average yield (for a given number of cycles), as discussed in the next Section 3.2.2.

In a next step, the aging parameters from each ÄKTA-DS in Table 4, were used to fit the two GPs (one per θ) from Eq. (22), having the accumulated PL and HF as input variables. The combination of the mechanistic model Eq. (4)–(12) and such GPs constitutes the hybrid model. By applying this model, the yield of the complete process can be simulated as a function of the cycle number for each of the considered cleaning procedure. However, since the complete cycle includes other steps, not considered

in the model, where the protein can be lost (e.g. in the washing phase), a correction has to be included in order to take this into account. In this work, all these effects are lumped in a correction factor of the maximum saturation capacity, q_{sat}^0 , which is fitted with data from one of the cycle studies (ÄKTA-DS4), by minimizing the error between the yield predictions and the experimental values. The so estimated correction factor is incorporated in the hybrid model, and applied to predict the yield of all the datasets.

In Fig. 5 (A-D), the predictions of the model (line) are compared with the experimental yields (dots) from ÄKTA-DS1 to 4, respectively. Please note that the yields of ÄKTA-DS1 were measured from pooled cycles and had to be interpolated, which leads to a smoother appearance. The hybrid model prediction is in line with the experimental values, with only ÄKTA-DS2 having an offset at low values of the yield, while reproducing the general decreasing trend. Once the hybrid model has been fitted, it provides a very strong tool in several model based activities such as simulation, control, design, and optimization.

3.2.2. Model based cleaning procedure optimization

At the process development stage, a lot of experimental effort is spent to obtain the optimal operating conditions and cleaning strategies. As already shown in Section 3.2.1, if the chosen HF is too mild, the operation of the process is limited by resin fouling. If the protocol is too harsh, ligand leaching and hydrolysis lead to a fast performance loss. In the following we show how to use the hybrid model developed above to find the optimal cleaning strategy. In particular, this is used to simulate process operations combining different PL and HF values. The objective of the optimization problem is to obtain the HF , for a given protein load, that maximizes the average yield over 100 cycles of resin life-time. Fig. 6 shows the response surface of the predicted average yield as a function of both PL and HF . It can be seen that for different protein loadings, the optimal HF changes accordingly. While for a PL of 45 g/(cm³ cycle) the optimal HF is 222, for 55 g/(cm³ cycle) it is 256. Thus, the hybrid model can help to design a CP that improves the performance of the process, even in conditions that were not tested experimentally.

It is worth noting that this approach could be very conveniently applied in the frame of a sequential optimization approach. This would include first solving the optimization problem and subsequently testing experimentally the suggested conditions. The new experimental data could be used to update the GP of the hybrid model and a new iteration of the optimization procedure could be performed. This concept is known as Bayesian optimization [41]. It is believed that supporting the development of the cleaning procedure by such modeling tools could indeed produce better results with significant savings and reduced experimental effort with respect to current empirical strategies.

3.2.3. Supervision of resin state

Yield is indeed the most important column performance indicator in the capture step and is typically used to decide when to change the resin [5]. As discussed above, the GP calculates the aging parameters of the column as a function of the accumulated values of PL and HF , which in turn could be used to estimate the process yield using the deterministic model of the column. However, due to differences in the lot-to-lot variability of impurities in the load material, column storage (Rathore et al., 2012), variations in the packing quality (Dorn et al., 2017), and other variations that can occur in manufacturing with respect to the resin lifetime studies conducted at the process development stage, the hybrid model could exhibit some deviations.

On the other hand, another potential information source is the UV signal of the loading and elution step for a given cycle. After

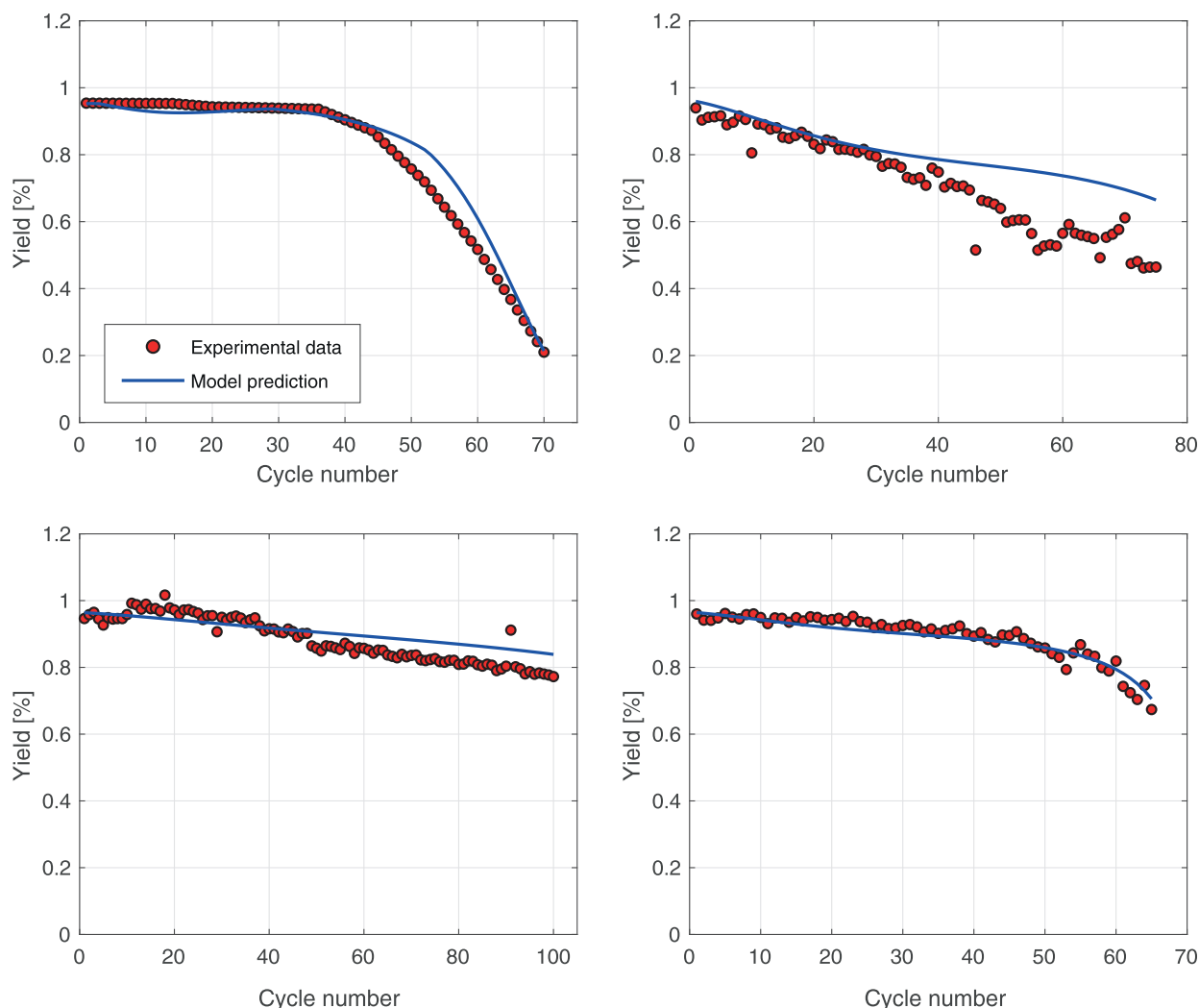


Fig. 5. External hybrid model predictions of the yield (line) as well as experimental reference values (dots) for ÄKTA-DS1 (A), ÄKTA-DS2 (B), ÄKTA-DS3 (C) and ÄKTA-DS4 (D)

UV sensor calibration, the on-line signal from the loading phase can be used to fit the aging parameters of the hybrid model and the signal from the elution phase can be used to roughly estimate the amount of eluted protein from the area under the peak, which in combination with the amount of protein fed to the column can be used to estimate the yield. Although this estimation uses real-time data and seems like the natural way of monitoring the process, it might be not very accurate, since the signal of the detector is not linear in the complete measurement range and operational differences can give rise to extreme outliers. However, this signal is usually not biased, and accounts for the unexpected variations that can take place in a manufacturing environment, as indicated above. Therefore, the hybrid model and the UV signal can be integrated into an extended Kalman Filter (KF) to create a soft sensor, predicting process yield in real-time. In this scheme, the KF allows to fuse information from both model and on-line sensor to obtain an improved estimation of the system state (yield). Once the state of the column has been corrected, it can be used in combination with the hybrid model to predict the yield and resin aging parameters of the following cycle.

In order to illustrate this approach, ÄKTA-DS1 is used as an example. The hybrid model uses the GP approach described in Eq. (23), since the KF requires a dynamic description of the

changes in the *aging parameters*. Here only BTCs from ÄKTA-DS1 are used to train the model, since it is expected that in manufacturing, process conditions such as the CP or loaded amount of protein remain fixed. Results are shown in Fig. 7 (A-B) compared with off-line HPLC measurements of the yields. Note that such yield values are not used by the KF and are shown here purely for comparison purposes. As can be seen in Fig. 7 (A), the estimation of the yield purely based on the on-line UV signal (green circles) is very noisy, even resulting in yields larger than 100%. The KF (blue curve) corrects these values using the predictions of the hybrid model (black curve). In spite of the mentioned offset, the KF shows good stability at the beginning of the operation and corrects the UV estimations. For lower values of the yield, the KF relies instead more on the information of the UV signal and corrects the significant offset of the hybrid model. Therefore, it can be concluded that the combination of those two information sources is important to improve the yield estimates in terms of both accuracy and robustness.

One of the most attractive feature of this tool is that it can also be used for process forecasting. Once the state of the system is corrected by the KF, the model alone can be used to forecast the yield in the following runs. It is then possible to predict in advance the yield decrease, so that maintenance activities, such

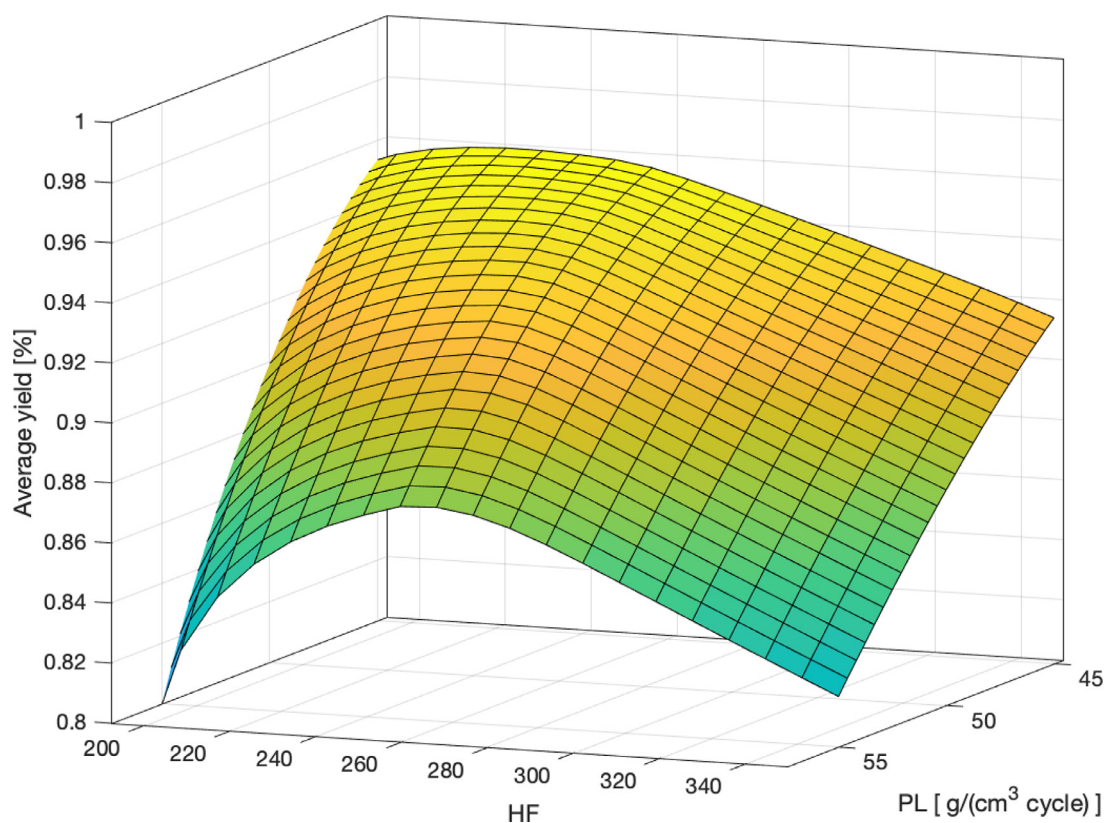


Fig. 6. Response surface of the average yield [%] over 100 cycles as a function of the harshness factor of the applied cleaning procedure and loaded amount of protein per cycle.

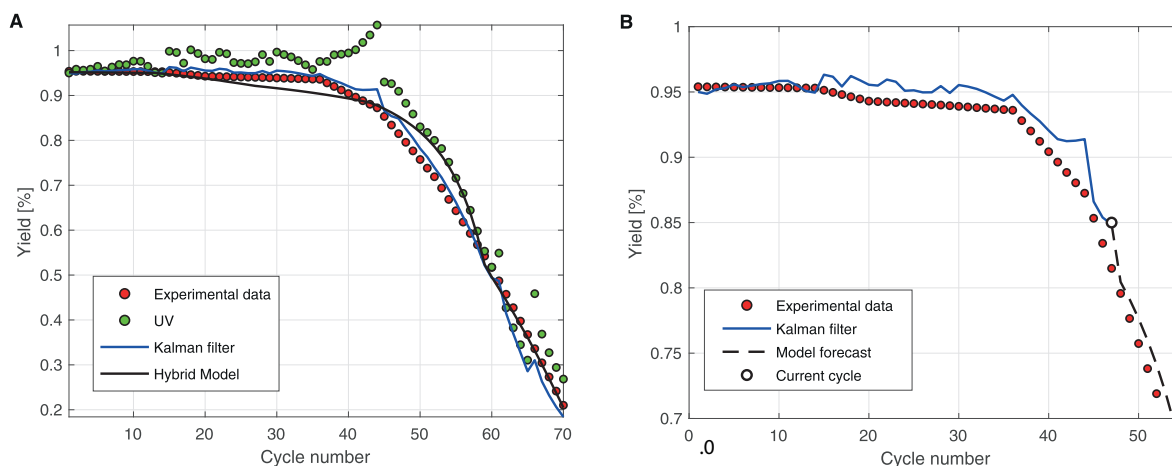


Fig. 7. (A) Kalman filter corrections: Yield from each cycle of ÄKTA-DS1 obtained from off-line experiments (red), UV signal (green), KF (blue) and hybrid model (black). (B) Model application for yield forecasting after cycle 40: Yield from each cycle of ÄKTA-DS1 obtained from off-line experiments (red), KF (blue), and model forecast (black), while the open circle indicates the current cycle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

as resin replacement, can be better planned. This is illustrated in Fig. 7 (B), where the KF is performed until run 46, after which only the model predictions are used. It can be seen that for the few following runs, the model anticipates the drop in the yield. It is easy to imagine that, with a bigger dataset to train the hybrid model on several complete lifetime studies, this tool can be designed and validated according to the Quality by Design paradigm. This could enable the use of CP parameters for on-line process control and optimization in the frame of a model predictive control, along the lines already explored in the area of continuous simulated chromatography (SMB) for chiral purification applications [42].

4. Conclusion and outlook

In this work, the important problem of resin aging and replacement is considered. In particular, two model approaches were developed to describe the complex phenomena of resin aging, understand the underlying aging mechanisms, optimize the cleaning procedure during process development and supervise the column quality during manufacturing towards predictive maintenance.

Each of the approaches showed advantages and limitations. The statistical approach provides a more holistic overview, since it takes several process parameters and chromatographic phases into

account to supervise the column quality or performance indicators such as yield, DBC or impurities. Additionally, it neither requires process knowledge nor ad-hoc experimental data for model tuning. However, it was observed that the approach shows limitations in extrapolating and, hence, this approach might be especially useful in manufacturing, where sufficient historical data of fixed operation conditions are available. Furthermore, it cannot provide predictions into the future.

The hybrid approach, on the other hand, exhibited an increased extrapolation capability due to the deterministic backbone, while leaving the explanation of the complex phenomena of resin aging to the statistical based components. A clear advantage of this procedure is the possibility to understand and describe the aging dependence on the used CP, which is important for model based CP optimization. Furthermore, this model can be used for forecasting the performance parameters, and can be updated on-line with the UV signal of the loading and elution phase. On the other side, this approach requires several data sets for tuning the model adjustable parameters and is more complex to develop. Additionally, it is limited to the prediction of yield, which however is the most important column performance indicator [5]. Here we see the potential application in both process development and manufacturing.

To apply such tools in real manufacturing conditions, additional elaborations and suitable data sets are required for their validation. However, this study allows concluding that statistical (PCA and PLS) and hybrid models could play an important and complementary role in process design and development as well as in manufacturing of biologics. In the case of the protein A capture step in the production of monoclonal antibodies, we have shown in this work at the laboratory scale, that the approaches can greatly extend the resin lifetime, which comes along with decreased cost of goods, reduced experimental effort, increased understanding, and process robustness.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

Fabian Feidl: Conceptualization. **Martin F. Luna:** Methodology. **Matevz Podobnik:** Formal analysis. **Sebastian Vogg:** Investigation. **James Angelo:** Conceptualization. **Kevin Potter:** Investigation. **Elenore Wiggin:** Investigation. **Xuankuo Xu:** Supervision. **Sanchayita Ghose:** Resources. **Zheng Jian Li:** Project administration. **Massimo Morbidelli:** Supervision. **Alessandro Butté:** Supervision.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.chroma.2020.461261](https://doi.org/10.1016/j.chroma.2020.461261).

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